TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	70.4 ± 2.91
Correlation Stability	86.0 ± 3.32
Distribution Consistency	78.9 ± 2.07
Entropy Preservation	82.5 ± 4.96
Mean Z Score Drift	0.6 ± 0.11
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.1 ± 0.30
Data Density	100.0 ± 0.00
Coverage Score	94.8 ± 1.99
Feature Accuracy	64.2 ± 5.74
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	6.326	+2.481	64.5%	Drift
Glucose	120.895	158.596	+37.702	31.2%	Drift
BloodPressure	69.105	82.702	+13.596	19.7%	Warn
SkinThickness	20.536	9.023	-11.514	56.1%	Drift
Insulin	79.799	68.585	-11.215	14.1%	Warn
BMI	31.993	40.286	+8.294	25.9%	Drift
DiabetesPedigreeFunction	0.472	0.541	+0.069	14.7%	Warn
Age	33.241	43.635	+10.394	31.3%	Drift
Outcome	0.349	0.581	+0.232	66.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	70.4 ± 2.91
Correlation Stability (*2)	86.0 ± 3.32
Distribution Consistency (*3)	78.9 ± 2.07
Entropy Preservation (*4)	82.5 ± 4.96
Mean Z Score Drift (*5)	0.6 ± 0.11
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.1 ± 0.30
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	94.8 ± 1.99
Feature Accuracy (*10)	64.2 ± 5.74

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.7311688311688311	0.7500000000000001	-0.00454545454545452	
F1	0.5998259463389416	0.4899683020960097	0.5893777422633301	-0.010448204075611534	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7266233766233766	0.7409090909090911	0.7337662337662337	0.00714285714285714	
F1	0.5913333771616245	0.5983246122353585	0.5921854694945489	0.0008520923329246255	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.675974025974026	0.6811688311688312	-0.08246753246753248	*
F1	0.6399852354635025	0.2058161404262094737	0.2584744309842633737	-0.381510804479239	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7116883116883117	0.6772727272727272	0.7168831168831169	0.005194805194805195	
F1	0.5705230392801266	0.5196940853929454	0.5816102810389012	0.01108724175877448	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.7129870129870131	0.7519480519480519	-0.004545454545454534	
F1	0.5733680734057942	0.3733042282947215	0.5425971884666131	-0.030770884939181244	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7506493506493508	0.7493506493506493	1.1102230246251566e-17	
F1	0.6076604413342895	0.5997511467284111	0.5956348440149173	-0.012025597319372494	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7487012987012986	0.7474025974025974	0.7694805194805194	0.020779220779220797	*
F1	0.6060826343485018	0.5846055791083289	0.6291205827637165	0.023037948415214716	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7461038961038961	0.7545454545454545	0.7493506493506493	0.0032467532467532756	
F1	0.6157553974310607	0.6106843794920124	0.6073657582245076	-0.008389639206553096	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7714285714285714	0.7305194805194806	0.7512987012987014	-0.020129870129870144	
F1	0.6264805246763366	0.5277662417412297	0.5660361421122634	-0.060444382564073175	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6896103896103896	0.6415584415584416	0.7019480519480519	0.012337662337662347	
F1	0.4682626672862387	0.4447711777097266	0.5441129272509522	0.07585025996471353	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	14.0	199.0	122.0	0.0	0.0	67.05997219996301	0.078	80.0	1.0	1.0
2	0.0	199.0	122.0	0.0	0.0	67.1	2.352120386184892	21.0	0.0	1.0
3	1.0	198.0	112.0	0.0	0.0	32.26778866324287	0.511889482717149	76.0	1.0	1.0
4	0.0	199.0	122.0	0.0	0.0	67.1	0.078	21.0	0.0	1.0
5	14.0	199.0	121.0	63.0	0.0	67.1	2.350473670795693	21.0	1.0	1.0
6	4.0	138.0	78.0	0.0	0.0	35.38368155853997	0.418633279012506	30.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	114.0	58.0	0.0	0.0	41.092184104366698	0.189460747892184	43.0	1.0	2.0
2	16.0	198.0	0.0	0.0	0.0	65.1397906406235	0.078	81.0	1.0	2.0
3	17.0	0.0	0.0	0.0	0.0	0.0	0.078	81.0	1.0	2.0
4	0.0	199.0	122.0	99.0	846.0	67.1	2.42	81.0	1.0	2.0
5	0.0	0.0	0.0	99.0	0.0	67.1	0.078	21.0	1.0	2.0
6	17.0	199.0	0.0	0.0	0.0	67.1	0.078	81.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	109.0	68.0	0.0	0.0	36.4048189796886	0.41265235257280	53.0	1.0	3.0
						6	924			
2	0.0	198.0	122.0	0.0	0.0	0.0	2.42	72.0	0.0	3.0
3	0.0	105.0	86.0	34.0	159.0	32.9032773047848	0.74427691167233	21.0	0.0	3.0
							42			
4	0.0	198.0	122.0	0.0	0.0	0.0	2.42	72.0	1.0	3.0
5	0.0	145.0	75.0	39.0	0.0	28.1792832574776	0.86141538715912	24.0	1.0	3.0
						07	02			
6	0.0	198.0	122.0	99.0	846.0	0.0	2.42	21.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	16.0	192.0	75.0	0.0	0.0	34.7358636423844	0.08738837806090	71.0	1.0	4.0
						16	491			
2	0.0	199.0	0.0	0.0	0.0	0.0	0.084	21.0	0.0	4.0
3	0.0	199.0	0.0	0.0	0.0	0.0	0.084	21.0	1.0	4.0
4	0.0	199.0	0.0	0.0	0.0	67.1	0.084	21.0	1.0	4.0
5	16.0	199.0	114.0	99.0	0.0	0.0	0.084	72.0	1.0	4.0
6	4.0	153.0	81.0	24.0	134.0	36.0274230573865	0.46300597142433	30.0	0.0	4.0
						7	164			
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	92.0	60.0	36.0	0.0	38.08179126964774	0.6342299284821531	45.0	1.0	5.0
2	0.0	199.0	122.0	0.0	0.0	67.09992870438032	0.078	81.0	0.0	5.0
3	0.0	93.0	81.0	43.0	178.0	39.011810117804686	0.6197018261208873	21.0	0.0	5.0
4	0.0	0.0	122.0	99.0	846.0	67.1	2.42	21.0	1.0	5.0
5	17.0	199.0	122.0	99.0	0.0	67.1	0.10131819306614089	81.0	1.0	5.0
6	5.0	151.0	84.0	24.0	107.0	35.777740080119341	0.4589353561437138	32.0	0.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	0.0	122.0	0.0	0.0	0.0	2.42	21.0	0.0	6.0
2	0.0	0.0	122.0	0.0	0.0	67.1	2.42	21.0	1.0	6.0
3	0.0	92.0	82.0	37.0	130.0	36.845887304417944	0.7091719054539514	21.0	0.0	6.0
4	0.0	199.0	122.0	0.0	846.0	67.1	2.42	21.0	1.0	6.0
5	3.0	118.0	65.0	31.0	192.0	33.933411105932246	0.7234594210709246	23.0	0.0	6.0
6	0.0	0.0	122.0	99.0	0.0	67.1	2.42	21.0	0.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	199.0	0.0	0.0	0.0	0.0	0.078	21.0	0.0	7.0
2	15.0	199.0	0.0	0.0	0.0	0.0	2.42	81.0	1.0	7.0
3	15.0	199.0	0.0	0.0	0.0	0.0	0.078	81.0	1.0	7.0
4	0.0	199.0	122.0	63.0	846.0	0.0	0.078	21.0	1.0	7.0
5	0.0	199.0	122.0	0.0	845.0	0.0	0.078	21.0	0.0	7.0
6	0.0	199.0	122.0	63.0	846.0	67.1	0.07803090884974932	21.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	0.0	122.0	63.0	846.0	0.0	0.078	21.0	0.0	8.0
2	0.0	0.0	122.0	0.0	0.0	67.1	2.42	21.0	1.0	8.0
3	0.0	105.0	5.0	0.0	0.0	38.638426216672954	0.5975805374073141	67.0	1.0	8.0
4	0.0	0.0	122.0	63.0	846.0	67.1	2.42	21.0	1.0	8.0
5	1.0	137.0	73.0	33.0	0.0	28.298407437962336	0.918236781070243	22.0	0.0	8.0
6	17.0	0.0	122.0	0.0	0.0	67.1	2.4111920015602033	77.0	0.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	9.0	124.0	25.0	0.0	0.0	65.00433906842932	0.08001991679023465	75.0	1.0	9.0
2	3.0	50.0	56.0	0.0	0.0	32.450048664376006	0.4579097679970368 7	22.0	0.0	9.0
3	0.0	0.0	122.0	0.0	0.0	0.0	0.078	21.0	0.0	9.0
4	0.0	0.0	122.0	99.0	846.0	67.1	2.329	21.0	1.0	9.0
5	0.0	143.0	70.0	36.0	0.0	28.20098224604229	0.7222392086044467	23.0	1.0	9.0
6	17.0	199.0	122.0	0.0	0.0	67.1	0.078	81.0	1.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	15.0	199.0	59.0	99.0	846.0	67.1	2.329	21.0	1.0	10.0
2	2.0	43.0	44.0	26.0	0.0	32.1202517596282	0.6381896527283432	21.0	0.0	10.0
3	0.0	199.0	0.0	0.0	0.0	67.1	0.078	70.0	1.0	10.0
4	0.0	199.0	122.0	99.0	0.0	0.0	0.078	70.0	1.0	10.0
5	0.0	141.0	77.0	38.0	0.0	29.233842620394793	0.8975294642505816	23.0	1.0	10.0
6	15.0	199.0	122.0	0.0	0.0	67.1	0.078	70.0	1.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan